

The Propagated Action Potential of Hodgkin and Huxley at Their 1952 Constants: A Computer-Assisted Existence Proof

Chase Hendrick, Independent Researcher (ORCID 0009-0002-9754-6087)

Abstract

In 1952 Hodgkin and Huxley computed their propagated action potential by hand. They shot in the conduction speed on their travelling-wave equation (J. Physiol. 117, eq. (31)), and noted that the solution "goes off towards either +infinity or -infinity" on the two sides of the speed.

The existence proofs that followed do not treat that equation. Hastings (1976) and Carpenter (1977) slow or speed the gating variables by a small parameter.

We prove that the unmodified equation has a pulse, an orbit homoclinic to rest, at 18.5 C and at 6.3 C. The rate functions and constants are those Hodgkin and Huxley printed, including the leak potential 10.613 mV. The proof is computer-assisted, in ball arithmetic. The speed parameter K lies in an interval of width $3e-45$ at 18.5 C and of width $2.8e-61$ at 6.3 C.

For the fibre constants of their p. 528, the conduction speed begins

18.73188824788048354046831343329624387695575077 m/s

at 18.5 C, against the 18.8 m/s they computed, and

12.31375672016229859330140853674732362368394751453213155988245 m/s

at 6.3 C. Every digit shown is proved.

The orbit leaves rest along its one-dimensional unstable manifold. A validated Taylor integrator carries a whole interval of speeds through the spike. An isolating block with a cone condition around rest, and a Wazewski-type shooting argument, closes it. [Computer-assisted.]

1. Introduction

The question. A travelling wave $V(x, t) = V(t - x/\theta)$ of the Hodgkin-Huxley cable equation solves eq. (31) of Hodgkin and Huxley (1952, p. 524),

$$\begin{aligned} d^2V/dt^2 &= K \{ dV/dt + (1/C_M) [g_K n^4 (V - V_K) + g_{Na} m^3 h (V - V_{Na}) + g_l (V - V_l)] \}, \\ K &= 2 R_2 \theta^2 C_M / a, \end{aligned}$$

together with their equations for m , n , h . A propagated action potential is a solution that starts and ends at rest. Hodgkin and Huxley found it numerically, at $K = 10.47$ /ms and 18.5 C (their p. 528). **Is there a proof that it exists, at their own constants?**

What was known (details in Section 6). Hastings (1976) proved existence for a class of Hodgkin-Huxley-type systems in which n and h are slowed by a factor ϵ , "for ϵ sufficiently

small”, and wrote that ”it is not clear that our results apply to the original HODGKIN-HUXLEY system” (p. 230). Carpenter (1977) proved existence for a generalized Hodgkin-Huxley system under abstract hypotheses, with n and h slowed by ϵ and m sped up by $1/\delta$, ”for small ϵ ” (Theorem 3.4) and ”all small δ ” (Theorem 4.2), and showed that the pulse is lost when either parameter is too large (Theorem 5.1(B)). Neither treats $\epsilon = \delta = 1$ at the 1952 functions. Computer-assisted proofs of travelling pulses exist for the FitzHugh-Nagumo equation (Arioli and Koch 2015, and others), not, as far as we found, for Hodgkin-Huxley.

What we prove. Theorems 1 and 2 (Section 3): at 18.5 C and at 6.3 C, with the 1952 rate functions and constants and the printed leak potential, the travelling-wave system has a pulse, with the speed parameter K in an explicit interval of width $3e-45$ at 18.5 C and $2.8e-61$ at 6.3 C. [Computer-assisted.] The same holds at 18.5 C with the leak potential that makes the resting current exactly zero (Remark 1). [Computer-assisted.]

Earlier work, as far as the search reached. The searches summarized in Appendix C (2026-09-25 to 2026-09-27) are recorded there. They are not a claim of priority. Two papers could not be obtained in full: Hastings (1976), of which we read pp. 229-230 of its 29 pages, and Foote and Chen, ”Traveling wave properties of the Hodgkin-Huxley equations”, Chinese J. Math. 9 (1981) 1-23, which we could not read at all. zbMATH Open has no review of either (Zbl 0374.35004, Zbl 0472.35048), and MathSciNet was not reachable. Carpenter (1977) was read in full.

What the proof rests on. Besides the computations, the proof uses only Hodgkin and Huxley's equations and constants (1952, pp. 519-528 and Table 3, read in a scan of the paper) and standard facts about ordinary differential equations, cited from one textbook that was read for the purpose (Teschl 2012: extension of solutions, Corollary 2.15; the flow is continuous on its open domain, Theorem 6.1; orbits in a compact set are complete and have nonempty, compact, invariant limit sets, Lemmas 6.3, 6.5 and 6.6; the local unstable manifold of a hyperbolic equilibrium, Theorems 9.4 and 9.5). Every other step is proved here (Section 4 and Appendices A and B). No step of the proof depends on Hastings (1976), Carpenter (1977), Foote and Chen (1981) or Arioli and Koch (2015); they bear only on the paragraph on earlier work and on the comparison of methods. The computations rest on the correctness of Arb (Johansson 2017), in FLINT, through python-flint 0.9.0, and, for the independent re-check of the closing block, of mpmath 1.3.0.

Methods and their sources. The integrator is a Lohner-type method (Lohner 1988): Taylor enclosures of the flow with the wrapping effect controlled by moving coordinates. The closing argument is a topological shooting argument in the tradition of Wazewski (1947) and of Conley's isolating blocks (Conley 1975, 1978), the approach Carpenter (1977) used for nerve impulse equations with small parameters. The blocks are checked through cone conditions, a quadratic form that increases along the flow, as in Zgliczynski (2009). Every lemma the proof uses is proved here from these ideas, so the citations credit the methods and are not premises.

2. The equations

We use the modern sign convention $u = -V$ (depolarization, mV), t in ms, $C_M = 1 \text{ uF/cm}^2$. With $u = -V$, the reversal potentials rewritten in the new sign ($V_{Na} = -115$, $V_K = 12$ and $V_l = -10.613$ mV become 115, -12 and $E_l = 10.613$ mV) and the rate functions written in u , eq. (31) keeps its form:

$$\begin{aligned} u' &= K (u' + I(u, m, n, h)), \quad I = 120 m^3 h (u - 115) + 36 n^4 (u + 12) + 0.3 (u - E_l), \\ x' &= \phi (\alpha_x(u) (1 - x) - \beta_x(u) x), \quad x = m, n, h, \quad \phi = 3^{((T - 6.3)/10)}, \end{aligned}$$

with $\alpha_m = \Psi((25 - u)/10)$, $\beta_m = 4 e^{-(u/18)}$, $\alpha_n = \Psi((10 - u)/10)/10$, $\beta_n = e^{-(u/80)}/8$, $\alpha_h = 0.07 e^{-(u/20)}$, $\beta_h = 1/(e^{((30 - u)/10)} + 1)$, $\Psi(x) = x/(e^x - 1)$ ($\Psi(0) = 1$), and $E_1 = 10.613$ mV, Hodgkin and Huxley's $V_1 = -10.613$ mV (Table 3) in our convention. The state is $y = (u, u', m, n, h)$ in $\mathbb{R}^5$, and we write the system as $y' = f(y, K)$; f is real analytic in (y, K) . The speed is $\theta = \sqrt{K a / (2 R_2 C_M)}$, with $a = 238$ μm and $R_2 = 35.4$ Ωcm for the fibre of their p. 528.

With the printed E_1 the resting current is not exactly zero at $u = 0$ (Table 3's footnote says the value was chosen to make it zero; with the printed rate functions the exact value is 10.5989...). Rest is then the equilibrium $y^* = (u^*, 0, m_{\text{inf}}(u^*), n_{\text{inf}}(u^*), h_{\text{inf}}(u^*))$ with

$u^* = 0.0036206688079425688368876905420\dots$ mV [computer-assisted: Lemma B.1; the only
zero of the
resting current within $1e-3$ mV of
this value]

and it depends neither on T nor on K . A pulse is a non-constant solution with $y(t) \rightarrow y^*$ as $t \rightarrow \pm\infty$.

The programs integrate these equations. `test_field.py` transcribes eq. (31) and Table 3 in the 1952 sign convention, with $V = -u$, $V_{\text{Na}} = -115$, $V_K = 12$ and $V_1 = -E_1$, and compares that transcription with `hhjet6.vfield`. At 64 states, at both temperatures and both leak potentials, including the two values where Ψ is singular, the largest relative difference is $5.22e-60$. Changing V_K to -12 in the transcription alone makes the two disagree, so the comparison is not vacuous. The neural-field reduction is a different paper and is not checked here.

3. Results

Theorem 1 (18.5 C). [Computer-assisted.] Let $T = 18.5$ C. For some K^* in (K_1, K_2) , with K_1 and K_2 the exact binary fractions of `data/pulse_proof_18.5_E10.613_config.json`,

$K_1 = 10.4380510601011236922764862383185791218586697704783\dots$, $K_2 - K_1 = 3.000e-45$ (to 4 digits),

the system of Section 2 has a pulse, and $\max u > 90.57$ mV along it. The corresponding speed lies in

(18.73188824788048354046831343329624387695575077276,
18.73188824788048354046831343329624387695575077548) m/s.

Theorem 2 (6.3 C). [Computer-assisted.] Let $T = 6.3$ C. For some K^* in (K_1, K_2) , with K_1 and K_2 the exact binary fractions of `data/pulse_proof_6.3_E10.613_config.json`,

$K_1 = 4.51063243827085108340326514573481715152474867970496\dots$, $K_2 - K_1 = 2.800e-61$ (to 4 digits),

the system of Section 2 has a pulse, and $\max u > 102.98$ mV along it. The corresponding speed lies in

(12.313756720162298593301408536747323623683947514532131559882457804,
12.313756720162298593301408536747323623683947514532131559882458189) m/s.

In both theorems the pulse leaves rest along a branch of the one-dimensional unstable manifold of y^* , the branch through the exit set E of Lemma 1, on which $u > u^*$ where it leaves the box B (the programs check $u - u^* > 0$ on all of E), and returns to y^* inside the block B_0 of Section 4.

Remark 1 (the zero-current leak potential). [Computer-assisted.] With $E_{-1} = 10.5989209693916785\dots$ (the value that makes the resting current zero at $u = 0$, so that $u^* = 0$) Theorem 1 holds with $K_2 - K_1 = 3.000e-45$ (to 4 digits) and the speed in (18.7321608143889 0211377538515402816936801773373586, 18.73216081438890211377538515402816936801773373858) m/s. The printed E_{-1} moves the speed by $2.7e-4$ m/s (to 2 digits).

Remark 2 (numerical, not proved). High-precision multiple shooting, at the printed leak potential, gives

$$K^*(18.5 \text{ C}) = 10.43805106010112369227648623831857912185866977197832623694577177829239$$

$$K^*(6.3 \text{ C}) = 4.510632438270851083403265145734817151524748679704964534920325310959260$$

each inside the interval $[K_1, K_2]$ of its theorem. Hodgkin and Huxley's $K = 10.47$ /ms is 0.3 per cent higher; with the same fibre constants it gives 18.76 m/s, which they report as 18.8 m/s. The measured speed in that fibre was 21.2 m/s. As rounding intervals of those printed values, $[10.465, 10.475]$ /ms, $[18.75, 18.85]$ m/s and $[21.15, 21.25]$ m/s are each disjoint from the interval of Theorem 1.

Not claimed. Uniqueness of the pulse or of K^* ; stability; other temperatures; the slow pulse that Huxley (1959) and later authors found numerically.

4. The proof

The proof has computed hypotheses (H2) to (H5) and an argument that uses only them and the facts from Teschl (2012) listed in Section 1. All computations are in ball arithmetic (FLINT/Arb through python-flint 0.9.0) at 256 bits (128 bits for enclosures that are only widened); each program stops on a failed check. Appendix A proves that the integrator's enclosures are enclosures, and Appendix B the interval lemmas the checks use.

For an invertible matrix P we use the coordinates $z = P (y - y^*)$. Since $f(y^*, K) = 0$, for y in a convex set Q that contains y^* ,

$$z' = A\text{-bar } z, \quad A\text{-bar} = \int_0^1 P Df(y^* + s (y - y^*), K) P^{-1} ds, \quad (4.1)$$

an average of the matrices $P Df(x, K) P^{-1}$ over x in Q . We write $\text{sym } S = (S + S^T)/2$.

Lemma 0 (a cone condition gives the inertia). Let A be a real $n \times n$ matrix and $D = \text{diag}(1, -1, \dots, -1)$. If $D A + A^T D$ is positive definite, then A has one simple real eigenvalue $\lambda_u > 0$, with an eigenvector v such that $v^T D v > 0$, and $n - 1$ eigenvalues with negative real part.

Proof. If $A v = \lambda v$ with v complex and nonzero, then $v^* (D A + A^T D) v = 2 \text{Re}(\lambda) v^* D v > 0$, so no eigenvalue is on the imaginary axis. Let E_+ and E_- be the real invariant subspaces of A belonging to the eigenvalues with positive and with negative real part; $\mathbb{R}^n$ is their direct sum. For x in $E_- \setminus \{0\}$, $x(t) = e^{\hat{t}A} x$ tends to 0 as $t \rightarrow +\infty$ and $d/dt (x(t)^T D x(t)) = x(t)^T (D A + A^T D) x(t) > 0$, so $x^T D x < 0$; likewise $x^T D x > 0$ on $E_+ \setminus \{0\}$, with $t \rightarrow -\infty$. Every 2-dimensional subspace contains a nonzero x with $x_{-1} = 0$, where $x^T D x \leq 0$, so $\dim E_+ \leq 1$; and E_- does not contain e_{-1} , so $\dim E_- \leq n - 1$. Hence $\dim E_+ = 1$: E_+ is spanned by a real eigenvector v of a real eigenvalue $\lambda_u > 0$ of algebraic multiplicity one, and $v^T D v > 0$. QED

(H2) Where the unstable manifold leaves a small box (`certify_rest_wave.lemma_B`). In $z = T_B (y - y^*)$, with T_B an exact binary matrix that approximately diagonalizes $Df(y^*)$

(unstable, fast real, complex pair, slow real), let

$$B = \{|z_1| \leq r_B, |z_2| \leq s_2, z_3^2 + z_4^2 \leq s_3^2, |z_5| \leq s_5\},$$

with $r_B = 1e-25$ (18.5 C) or $1e-32$ (6.3 C) and s_j of order r_B^2 (the values are in the configuration files). Checked in ball arithmetic over $B \times [K_1, K_2]$: (i) $D A + A^T D$ is positive definite for every A in an interval matrix enclosing $T_B Df(y, K) T_B^{-1}$, $D = \text{diag}(1, -1, -1, -1, -1)$ (Gershgorin's bound, Lemma B.5); (ii) the stable faces are strictly inflowing: $z_2 z_2' < 0$ where $|z_2| = s_2$, $z_3 z_3' + z_4 z_4' < 0$ where $z_3^2 + z_4^2 = s_3^2$, and $z_5 z_5' < 0$ where $|z_5| = s_5$, at every point of B ; (iii) $z_1' > 0$ on the exit face $E = B$ intersected with $\{z_1 = r_B\}$.

Write $L = z_1^2 - (z_2^2 + \dots + z_5^2)$. By (i), (4.1) with $P = T_B$ and $Q = B$, and the concavity in A of the smallest eigenvalue of $D A + A^T D$, $dL/dt = z^T (D A\text{-bar} + A\text{-bar}^T D) z > 0$ at every point of B other than y^* .

(H1) Rest is hyperbolic, with a one-dimensional unstable manifold. By Lemma 0 applied to $T_B Df(y^*, K) T_B^{-1}$ (y^* lies in B), for every K in $[K_1, K_2]$. As an independent check, `certify_rest_wave.lemma_A` encloses the characteristic polynomial of $Df(y^*, K)$ over the K interval, shows that it has one simple real root λ_u in an interval, and checks the Routh-Hurwitz inequalities for the quotient (Lemma B.4); the proof does not use this check.

Lemma 1. Assume (H2). For every K in $[K_1, K_2]$:

- (a) the unstable manifold of y^* has a branch, the orbit Γ_K of a solution x_K with $x_K(t) \rightarrow y^*$ as $t \rightarrow -\infty$ and $z_1 > 0$ on it near y^* , that stays in B until a first time t_e , at which it leaves B through a point $p(K)$ of E ;
- (b) $p(K)$ is the only point of E whose backward orbit stays in B , and $K \rightarrow p(K)$ is continuous on $[K_1, K_2]$.

Proof. (a) By Lemma 0, y^* is hyperbolic with a one-dimensional unstable space spanned by v , with $v^T D v > 0$ in z coordinates, so v_1 is not 0. By Teschl's Theorems 9.4 and 9.5 the points near y^* whose backward orbits stay near y^* form a C^1 curve through y^* , tangent to v , that consists of y^* and two orbits; near y^* , z_1 has opposite signs on them, and $L > 0$ on them (tangency and $v^T D v > 0$). Let Γ_K be the orbit with $z_1 > 0$ and x_K a solution on it. The point $z = 0$ is interior to B , so $x_K(t)$ lies in B for all t below some t_0 . While x_K is in B , L increases, so $L > 0$ and z_1 cannot vanish: $z_1 > 0$. Let $t_e = \sup \{t : x_K(s) \text{ in } B \text{ for all } s \leq t\}$. If t_e were infinite, the forward orbit would lie in the compact set B ; by Teschl's Lemmas 6.3, 6.5 and 6.6 its omega-limit set would be nonempty, compact, invariant and inside B , and L , increasing and bounded along the orbit, would be constant on it; since $dL/dt > 0$ on $B \setminus \{y^*\}$, the omega-limit set would be $\{y^*\}$, and $L(x_K(t))$ would increase to $L(y^*) = 0$ from positive values, which is impossible. So t_e is finite, $x_K(t_e)$ is on the boundary of B , and there are times $t > t_e$ arbitrarily close to t_e at which $x_K(t)$ is outside B . One of the constraints $g_j \leq 0$ that define B ($g = z_2^2 - s_2^2$, and so on) is violated at such times while $g_j(x_K(t_e)) = 0$, so $d g_j(x_K(t))/dt \geq 0$ at t_e . By (ii) this is impossible for the stable faces, so $x_K(t_e)$ is on a face $|z_1| = r_B$, and since $z_1 > 0$, $p(K) = x_K(t_e)$ is in E .

- (b) Let q be in E , with its backward orbit in B . By Teschl's Lemma 6.3, q is backward complete; by Lemmas 6.5 and 6.6 its alpha-limit set is nonempty, compact, invariant and in B , and L is constant on it, so as in (a) it is $\{y^*\}$. So the backward orbit tends to y^* : otherwise there are times $t_n \rightarrow -\infty$ at which it stays at distance at least some $\epsilon > 0$ from y^* , and since B is compact a subsequence converges to a point of the alpha-limit set other than

y^* . Then L decreases to 0 along it in backward time, hence $L > 0$ and $z_1 > 0$ on it, and by Teschl's Theorem 9.5 it eventually lies on the local unstable manifold, on the branch with $z_1 > 0$: q is on Γ_K . A point of Γ_K after $p(K)$ has a backward orbit through points just after $p(K)$, where $z_1 > r_B$ by (iii), so outside B . A point of Γ_K in E before $p(K)$ would, by (iii), be followed at once by points with $z_1 > r_B$, outside B , before t_e , contradicting the definition of t_e . So $q = p(K)$.

Continuity: let $K_n \rightarrow K$ in $[K_1, K_2]$. Since E is compact it suffices to show that every limit q of a subsequence of $p(K_n)$ is $p(K)$. Consider the system with K as a sixth variable, $K' = 0$, and its flow Φ , continuous on its open domain (Teschl, Theorem 6.1). For every $t \leq 0$ in the maximal interval of the solution through (q, K) , the points $\Phi(t, (p(K_n), K_n))$ are defined for large n (the domain is open) and lie in B , so their limit $\Phi(t, (q, K))$ lies in B (B is closed). By Teschl's Lemma 6.3 the solution through q is then defined for all $t \leq 0$, and its backward orbit lies in B . By (b), $q = p(K)$. QED

(H3) The closing block (block0.py). In $zeta = M(y - y^*)$, $M = \text{diag}(10, 7, 1, 1, 40)$ T (T an exact binary approximate inverse eigenbasis), let $B_0 = \{|zeta_1| \leq r, |zeta_s|_2 \leq \rho\}$, $zeta_s = (zeta_2, \dots, zeta_5)$, with $\rho = 0.8$, $r = 0.84$ (18.5 C) and $\rho = 0.6$, $r = 0.63$ (6.3 C), as binary floating-point numbers that the programs use exactly: $\rho = 0.8000000000000000444\dots$, $r = 0.8 \times 1.05$ rounded to $0.8400000000000000799\dots$, and $\rho = 0.5999999999999999777\dots$, $r = 0.6300000000000000044\dots$. Checked on a cover of B_0 by cells (Lemma B.3), with interval Cholesky factorizations (Lemma B.2), for every K in the interval: (C) $D A + A^T D$ is positive definite for every A in an interval matrix enclosing $M Df(x) M^{-1}$ over the cell; (E) on the cells that meet B_0 and $\{|zeta_1| \leq \rho\}$, $-\text{sym}(A_{ss}) - \mu I$ is positive definite, where μ is an upper bound of the Euclidean norm of the column A_{s1} over the cell; so $\lambda_{\max}(\text{sym } A_{ss}) + |A_{s1}|_2 < 0$ there.

Lemma 2. Assume (H3). While an orbit is in B_0 and not at y^* , $L = zeta_1^2 - |zeta_s|_2^2$ increases strictly; every boundary point of B_0 with $L \leq 0$ is a point of strict entrance; the cones $K+ = \{L > 0, zeta_1 > 0\}$ and $K- = \{L > 0, zeta_1 < 0\}$ cannot be left while the orbit stays in B_0 ; and an orbit that stays in B_0 for all $t \geq t_0$ tends to y^* .

Proof. Apply (4.1) with $P = M$ and $Q = B_0$, or $Q = B_0$ intersected with $\{|zeta_1| \leq \rho\}$; both are convex and contain y^* . The smallest eigenvalue of $D A + A^T D$ is a concave function of A and $\lambda_{\max}(\text{sym } A_{ss}) + |A_{s1}|_2$ is a convex one, so the bounds (C) and (E), which hold for every $M Df(x) M^{-1}$ with x in the region, hold for the averages $\bar{A}$. Then $dL/dt = zeta^T (D \bar{A} + \bar{A}^T D) zeta > 0$ for $zeta \neq 0$. At a boundary point with $L \leq 0$ we have $|zeta_s|_2 = \rho$ and $|zeta_1| \leq \rho$ (because $r > \rho$), and $(1/2) d|zeta_s|_2^2/dt = zeta_s^T (\bar{A}_{ss} zeta_s + \bar{A}_{s1} zeta_1) \leq (\lambda_{\max}(\text{sym } \bar{A}_{ss}) + |\bar{A}_{s1}|_2) \rho^2 < 0$, so the orbit is outside B_0 just before and inside just after. While the orbit is in B_0 , $L > 0$ persists and $zeta_1$ cannot vanish, so it keeps its sign. An orbit that stays in B_0 for $t \geq t_0$ has, by Teschl's Lemmas 6.3, 6.5 and 6.6, a nonempty invariant omega-limit set in B_0 on which L is constant, hence $\{y^*\}$; and the orbit tends to y^* , since otherwise times $t_n \rightarrow \infty$ at which it stays at distance at least some $\epsilon > 0$ from y^* would have, B_0 being compact, a subsequence converging to a point of the omega-limit set other than y^* . The face $|zeta_1| = r$ is not used. QED

(H4) The interval run (hh_prove_pulse.py interval). A Lohner-type integrator in the six variables (y, K) , $K' = 0$ (lohner6.py; Taylor jets of order 40 with derivatives in the initial point, hhjet6.py), carries a set containing $E \times [K_1, K_2]$ from $t = 0$ to $t = T_{\text{enter}}$ and encloses it, at T_{enter} , in the interior of B_0 . By Appendix A, for every (q, K) in $E \times [K_1, K_2]$ the solution

through (q, K) exists on $[0, T_enter]$ and its value at T_enter lies in $\text{int } B0$.

(H5) The endpoint runs (`hh_prove_pulse.py K1, K2`). For $K = K1$ (resp. $K2$) the set E is carried to T_enter , lies in $\text{int } B0$ there, and is carried further in steps of 2^{-7} ms, with the whole path of every step enclosed (Lemma A.4) and inside $\text{int } B0$, until the set lies in $K-$ (resp. $K+$).

Proof of Theorems 1 and 2 from (H2) to (H5). For K in $[K1, K2]$ let x_K be the solution with $x_K(0) = p(K)$. It lies on the unstable manifold, so $x_K(t) \rightarrow y^*$ as $t \rightarrow -\infty$. By Lemma 1(b) and the continuity of the flow of the six-variable system on its open domain (Teschl, Theorem 6.1), $K \rightarrow x_K(t)$ is continuous, uniformly for t in compact intervals on which the solutions exist. Let $S+$ (resp. $S-$) be the set of K in $[K1, K2]$ for which there is $t \geq T_enter$ with $x_K([T_enter, t])$ in $\text{int } B0$ and $x_K(t)$ in $K+$ (resp. $K-$). Both are open in $[K1, K2]$: the conditions are open and concern a compact time interval. They are disjoint (Lemma 2: a cone cannot be left while in $B0$, and an orbit in $K+$ never meets $K-$). $K2$ is in $S+$ and $K1$ in $S-$ (H5, since $p(K1)$ and $p(K2)$ are in E). Since $[K1, K2]$ is connected, some K^* is in neither. By (H4), $x_{\{K^*\}}(T_enter)$ is in $\text{int } B0$. If $x_{\{K^*\}}$ left $B0$, then at the first time t_e at which it reaches the boundary, either $L \leq 0$, and then by Lemma 2 the orbit enters $B0$ strictly there, so it was outside $B0$ just before t_e , which it was not; or $L > 0$, and then it was in $K+$ or $K-$ just before t_e while in $\text{int } B0$, so K^* would be in $S+$ or $S-$. Hence $x_{\{K^*\}}(t)$ stays in $B0$ for $t \geq T_enter$ (and exists for all such t , by Teschl's Lemma 6.3) and tends to y^* (Lemma 2). It is not constant. So it is a pulse. The speed bounds follow from $\theta = \sqrt{K a / (2 R_2 C_M)}$ in ball arithmetic, with the decimal ends rounded outward. The lower bound on $\max u$ is the lower end of an enclosure of u at a step end of the interval run. QED

What is not part of the proof. The numerical centre K^* (Remark 2), used only to place $[K1, K2]$; the choice of the weights, the radii, the matrices T_B , T and M , and T_enter ; floating-point step-size heuristics (they choose step lengths; every enclosure is checked); the characteristic-polynomial check of (H1).

5. Computations, controls and checks

Table 1: stages at 18.5 C (printed E_1), one process at a time, 256 bits, order 40. The times are the wall-clock seconds the certificates record (**secs**), not processor time.

stage	result	wall time
setup (H2, H3, the check of H1)	$\lambda_u = 10.89208\dots$; Lemma B at $r_B = 1e-25$; $z1' > 0$ and $u > u^*$ on the exit set; $B0$ certified on $1232 + 5916$ cells	seconds
interval (H4)	at $T_enter = 13.625$ ms: ζ_1 in $[-0.341, 0.341]$, $\text{abs}(\zeta_s) \leq 0.63603 < 0.8$	376 s
K1 (H5)	enters $K-$ at 13.6953125 ms, path in $\text{int } B0$	376 s
K2 (H5)	enters $K+$ at 13.6875 ms, path in $\text{int } B0$	375 s

stage	result	wall time
negative control: K interval shifted by 40 half-widths	zeta_1 about 13 at T_enter , outside B0: fails, as it must	373 s
negative control: α_m times $(1 + 1e-12 (u - u^*)^2)$	the whole set escapes below $u = -60$ mV at 6.72 ms: fails, as it must	390 s
negative controls in setup	a bracket above λ_u ; Lemma B faces 100 times thinner; B0 with radius x 1.5: all rejected	seconds

Table 2: the same at 6.3 C (printed E_1).

stage	result	wall time
setup (H2, H3, the check of H1)	$\lambda_u = 4.974030\dots$; Lemma B at $r_B = 1e-32$; $z1' > 0$ and $u > u^*$ on the exit set; B0 certified on 3590 + 1374 cells	seconds
interval (H4)	at $T_enter = 36.125$ ms: zeta_1 in $[-0.273, 0.273]$, $\text{abs}(\text{zeta_s}) \leq 0.45631 < 0.6$	852 s
K1 (H5)	enters K- at 36.2578125 ms, path in int B0	849 s
K2 (H5)	enters K+ at 36.234375 ms, path in int B0	852 s
negative control: K interval shifted by 40 half-widths	zeta_1 about 10 at T_enter , outside B0: fails, as it must	856 s
negative control: α_m times $(1 + 1e-12 (u - u^*)^2)$	the whole set escapes below $u = -60$ mV at 17.20 ms: fails, as it must	860 s
negative controls in setup	as at 18.5 C: all rejected	seconds

The model control multiplies α_m by $1 + 1e-12 (u - u^*)^2$, with u^* the rest-voltage ball rather than its midpoint, so the factor is 1 only at a single point of that ball. It re-runs the cone and inflow test. It does not compare linearizations, and it does not re-check $z1' > 0$. (H3) is not checked again; on B0, where $|u - u^*|$ is of order 1 mV, the factor changes the field by about $1e-12$. The control only has to fail, and the perturbation moves the pulse speed by far more than the width of the K interval. At 6.3 C the numerical centre has to be computed with a local error budget $1e-8$ times tighter than at 18.5 C: the budget is written for the growth rate at 18.5 C, and the looser centre differs from the tighter one by about $5e-59$, far more than the width of the K interval. The proof uses the tighter centre.

Independent re-check of (H3). `hh_block_check_iv.py` is a separate program: mpmath interval arithmetic at 113 bits, the Jacobian from hand-derived formulas, M^{-1} in exact rational arithmetic,

its own cover and Cholesky test, ϕ from the decimal temperature, and its own enclosure of the rest state by bisection, which it checks to lie within the window where Lemma B.1 shows that the zero is unique. It confirms (C) and (E) at both temperatures and rejects the enlarged block with the same depth limit as the check itself.

Tests. `test_field.py` compares the vector field of the programs with an independent transcription of the 1952 equations in Hodgkin and Huxley's own sign convention, at 64 states at both temperatures and both leak potentials, including the removable singularities at $u = 10$ and $u = 25$ and states next to them; the largest relative difference is $5.2e-60$, and the same transcription with $V_K = -12$ is caught. `test_lohner6.py`: the six-variable jets agree with an independent Picard implementation and with central differences in K ; Lohner enclosures through the upstroke, at orders 30 and 8, contain a high-precision reference solution computed with the same jets and a different step sequence (so this tests the enclosures, not the field); at order 8 with the remainder term dropped the enclosure misses it, as it must. `test_temperature.py` checks that ϕ is computed from the decimal temperature, so that $\phi = 1$ at 6.3 C; the binary number nearest 6.3, 6.2999999999999982 C, gives $\phi = 1 - 1.95e-17$, and the test rejects it.

The harness. Every certificate records the sha256 of the configuration, of the closing block and of the eight programs the proof runs, the python-flint version, and the ϕ and E_1 balls. The summary recomputes all of these, and each verdict from the certificate's own fields, and refuses a certificate that does not match, whose K is not the configuration's, or whose negative control failed for a reason other than its stated one. A summary control plants thirteen kinds of stale, foreign or self-contradictory certificate in a copy of the data and requires the summary to refuse each, and to accept the unaltered copy.

Consistency. At $T_{\text{enter}} = 13.625$ ms (18.5 C, printed E_1) the certificates enclose $\zeta_1(K_1)$ in $[-0.3032 \pm 1.03e-5]$ and $\zeta_1(K_2)$ in $[0.33917 \pm 8.72e-6]$.

6. Comparison with Carpenter (1977) and Hastings (1976)

Carpenter's Theorem 3.4 (p. 353) proves, under the abstract Hypotheses (3.1, CUBIC, H) and (3.3, HOM, H), that the reduced system (3.1, H), with $m = m_{\text{inf}}(V)$ and n, h multiplied by ϵ , has a homoclinic solution "for small $\epsilon > 0$ "; Theorem 4.2 (p. 357) restores m as a fast variable "for all small $\delta > 0$ "; and Theorem 5.1(B) (pp. 357-358) shows that the pulse is lost when ϵ or δ is too large. Her method, isolating blocks and a Wazewski-type shooting in the speed around a singular orbit, is the same kind of topological argument as ours, applied where the small parameters make the orbit computable by hand; ours applies it to a validated numerical orbit at $\epsilon = \delta = 1$. `zbMATH` lists her lecture notes "Nerve impulse equations" (Carpenter 1976) next to the 1977 paper; we have not read them. Hastings's theorem (pp. 229-230, read) needs n and h slowed by a small ϵ and hypotheses he did not verify for the 1952 functions.

7. Reproducibility

The companion repository is `ChaseHendrick/hh-pulse`. Checking release 1.0.1 archives the exact existence-proof programs, certificates and supplementary checking scripts at [doi:10.5281/zenodo.23028512](https://doi.org/10.5281/zenodo.23028512).

The programs are in `code/` (Apache-2.0) and need `python-flint 0.9.0`, `mpmath`, `numpy` and `scipy` (`code/requirements.txt`). From the folder of this paper:

```
python3 -m pip install -r code/requirements.txt
sh code/run.sh all          # or: tests, 18.5, 6.3, zero
```

`run.sh` runs `test_lohner6.py` and then, for each proof, from scratch and one process at a time: the numerical centre (`hp_pulse.py`), the block (`block0.py`), the configuration and the stages of `hh_prove_pulse.py` (setup, interval, K1, K2 and the two negative controls), the independent block check (`hh_block_check_iv.py`) and the summary, whose exit status is 0 if and only if every check passed and every negative control failed. The certificates are `data/pulse_proof_<T>_E110.613_*.json` (printed `E_1`) and `data/pulse_proof_18.5_*.json` (zero-current `E_1`), with the summaries `*_summary.txt`. The starting profiles `data/pulse_<T>.npz` are numerical initial guesses made by `pulse_bvp.py`.

Appendix A. Validated integration

The integrator works with sets $X = \bar{x} + C r_0 + B r$, r_0 in R_0 , r in R ($\bar{x}$ a point of R^6 , C and B matrices, R_0 and R boxes), in the variables (y, K) with $K' = 0$. Write $[X]$ for the interval hull of X and $F(W)$ for an interval enclosure of f over a box W . The Taylor coefficients $x_k(x_0)$ of the solution through x_0 ($x(t) = \sum x_k(x_0) t^k$) are computed by the standard recursion on truncated power series (automatic differentiation of the field) in ball arithmetic, so that for a box Q the computed ball x_k contains $x_k(x_0)$ for every x_0 in Q . The only function beyond the arithmetic operations and $\exp$ is Ψ ; near 0 it is $1/G$ with $G(x) = (e^x - 1)/x = \sum_n x^n/(n+1)!$, whose Taylor coefficients at a ball x_0 with $|x_0| \leq 1/2$ are summed to $n = 400$, with the tail bounded by twice the first omitted term (for $n \geq 2k + 2$ the ratio of consecutive terms of the k -th coefficient is at most $2|x_0|/(n+2) \leq 1/2$).

Lemma A.1 (a priori enclosure). Since $K' = 0$, the K component of a solution is constant, and the statement is about the five moving components y . Let $W = W_y \times W_K$ be a box in $R^5 \times R$ whose K component W_K contains the K component of $[X]$, let $F_y(W)$ be an interval enclosure of $f(y, K)$ over W , and suppose that $[X]_y + [0, h] F_y(W)$ is contained in the interior of W_y , where $[X]_y$ is the hull of the y components of X . Then for every (y_0, K) in X the solution of $y' = f(y, K)$, $y(0) = y_0$, exists on $[0, h]$ and lies in W_y there.

Proof. Fix (y_0, K) in X ; then K is in W_K . Let τ be the supremum of the t in $[0, h]$ such that the solution exists on $[0, t]$ and lies in W_y there. For $t < \tau$, $y(t) = y_0 + \int_0^t f(y(s), K) ds$ lies in $y_0 + t F_y(W)$ ($F_y(W)$ is a box, hence convex), a subset of the compact set $[X]_y + [0, h] F_y(W)$, which lies in $\text{int } W_y$. By Teschl's Corollary 2.15 the solution extends beyond τ , and by continuity it stays in $\text{int } W_y$ a little longer; so $\tau = h$, and the solution lies in W_y on $[0, h]$. QED

In the six variables the solution $(y(t), K)$ then lies in W on $[0, h]$; the program checks the inclusion for the five moving components only (for the endpoint runs, where K is a point, a six-dimensional interior would be empty).

Lemma A.2 (Lagrange remainder). Under Lemma A.1, for t in $[0, h]$ and each component i , $x_i(t) - \sum_{k \leq p} x_{i,k}(x_0) t^k$ lies in $t^{p+1} x_{i,p+1}$ for any box W' that contains the solution on $[0, h]$; if boxes $W'_1, \dots, W'_m$ contain it on subintervals that cover $[0, h]$, it lies in the hull of the $t^{p+1} x_{i,p+1}$.

Proof. Taylor's theorem with the Lagrange remainder for the real function x_i gives the remainder $x_i^{(p+1)}(\xi) t^{p+1}/(p+1)!$ with ξ in $(0, t)$, and $x_i^{(p+1)}(\xi)/(p+1)! = x_{i,p+1}(x(\xi))$

because the system is autonomous; $x(x_i)$ lies in W' (or in the W'_j of a subinterval containing x_i). QED

Lemma A.2 with a lower order in place of p and t in $[0, h]$ gives tighter enclosures of the path on $[0, h]$ or on subintervals, which are then used as W' .

Lemma A.3 (mean-value form and the new set). Let $\Phi(x_0) = \sum_{k \leq p} x_k(x_0) h^k$, $[J] = \sum_{k \leq p} h^k D x_k$, and let Rem be the remainder box of Lemma A.2 at $t = h$. With $y = \Phi(\bar{x}) + \text{Rem}$, $\bar{x}' = \text{mid}(y)$, $C' = \text{mid}([J] C)$, B' an invertible point matrix, $[B'^{-1}]$ an enclosure of its inverse, and

$$R' = [B'^{-1}](y - \bar{x}' + ([J] C - C') R_0) + ([B'^{-1}] [J] B) R,$$

the solution at time h from every x_0 in X lies in $X' = \bar{x}' + C' r_0 + B' r'$, r_0 in R_0 , r' in R' .

Proof. The point $\bar{x}$ lies in X , and so the segment from $\bar{x}$ to x_0 lies in the convex hull of X and in $[X]$: the boxes R_0 and R contain 0 at every step. R_0 is symmetric about 0 by construction and does not change; the initial R contains 0 by construction; and if R contains 0, so does R' , since $y - \bar{x}'$ contains 0 ($\bar{x}' = \text{mid}(y)$), $([J] C - C') R_0$ contains 0 (R_0 does), and $([B'^{-1}] [J] B) R$ contains 0 (R does). For each component, the mean value theorem on that segment gives $\Phi_i(x_0) = \Phi_i(\bar{x}) + \text{grad } \Phi_i(x_i) (x_0 - \bar{x})$ with $\text{grad } \Phi_i(x_i)$ in the i -th row of $[J]$. So $x(h) = \Phi(x_0) + \text{rem}$ with rem in Rem , and $x(h) - \bar{x}' - C' r_0 = (\Phi(\bar{x}) + \text{rem} - \bar{x}') + (J C - C') r_0 + J B r$ for a matrix J in $[J]$. Multiplying by B'^{-1} and enclosing each term gives $r' = B'^{-1} (x(h) - \bar{x}' - C' r_0)$ in R' . QED

In the program B' is the orthogonal factor of a QR factorization of $\text{mid}([J] B)$, and `arb_mat.inv` encloses its inverse; $\Phi(\bar{x})$ and the linear algebra are at 256 bits, $[J]$, W and Rem at 128 bits (wider balls, still enclosures).

Lemma A.4 (the path of a step). Under Lemma A.1, for every t in $[0, h]$ the solution lies in $\sum_{k \leq p} x_k [0, h]^k + [0, h]^{(p+1)} x_{p+1}$. *Proof.* Lemma A.2 with t in $[0, h]$ and the interval extension of each term. QED

In (H5) this box, mapped to zeta coordinates, is checked to lie in $\text{int } B_0$ for every step.

Appendix B. Interval lemmas

Lemma B.1 (interval Newton in one variable). Let g be C^1 on $X = [a, b]$, m in X , G' a closed interval that contains $g'(X)$ and not 0, and $N = m - g(m)/G'$ (interval arithmetic) contained in the interior of X . Then g has exactly one zero in X , and it lies in N .

Proof. g' has constant sign on X , so g has at most one zero there, and a zero x satisfies $x = m - g(m)/g'(x)$, in N , by the mean value theorem. Existence: say $G' = [c, d]$ with $c > 0$ and $g(m) > 0$ (the other cases are symmetric, and $g(m) = 0$ is trivial). Then $N = [m - g(m)/c, m - g(m)/d]$, and N in $\text{int } X$ gives $m - g(m)/c > a$, so $g(a) \leq g(m) - c(m - a) < 0 < g(m)$, and g has a zero in (a, m) . QED

`certify_rest_wave.rest_state` applies Lemma B.1 to the resting current $g(u) = I(u, m_{\text{inf}}(u), n_{\text{inf}}(u), h_{\text{inf}}(u))$ on the interval of radius $1e-3$ mV around a floating-point zero, and then intersects further Newton steps down to a radius of about $1e-73$.

Lemma B.2 (interval Cholesky). Let H be an interval matrix whose lower triangle contains the lower triangle of every symmetric matrix S in a set. If the Cholesky recursion, carried out in

interval arithmetic on the lower triangle of H , produces pivots whose intervals are positive, then every such S is positive definite.

Proof. Run the same recursion on S in exact arithmetic. By induction on the steps, each quantity it computes lies in the corresponding interval (inclusion isotonicity), so each pivot is positive and the recursion does not break down. It produces a real lower-triangular L with positive diagonal and $S = L L^T$, so S is positive definite. QED

In (H3)(C), $S = D A + A^T D$ for A in the interval matrix of the region, and the interval entries of H are evaluated from those of A , so they contain the entries of S . (H2)(i) uses Lemma B.5 instead.

Lemma B.3 (covers). `block0.cover_check` starts from the box $\{|\zeta_1| \leq r, |\zeta_j| \leq \rho, j = 2..5\}$, which contains B_0 (for the entrance check, from the same box with $|\zeta_1| \leq \rho$), and bisects coordinates in a fixed cycle. A cell is discarded only when a rigorous lower bound of $|\zeta_s|$ over it exceeds ρ , so that it misses B_0 , and it is accepted only when its check passes; a cell that still fails at the maximal depth ends the program with a failure. So the accepted cells cover B_0 (resp. B_0 intersected with $\{|\zeta_1| \leq \rho\}$).

Lemma B.4 (the check of (H1)). With λ_u enclosed in $[a, b]$ ($P(a) < 0 < P(b)$, $P' > 0$ on $[a, b]$), synthetic division of the enclosed coefficients of the characteristic polynomial P by $x - [a, b]$ encloses the coefficients of the quotient Q for every K and every root in $[a, b]$, and the remainder encloses 0. The Routh-Hurwitz inequalities for the quartic $Q/q_4 = x^4 + a_3 x^3 + a_2 x^2 + a_1 x + a_0$, namely $a_3, a_2, a_1, a_0 > 0$, $a_3 a_2 - a_1 > 0$ and $a_3 a_2 a_1 - a_1^2 - a_3^2 a_0 > 0$ (Teschl 2012, p. 72, eq. (3.45)), are checked on those enclosures. The proof does not use this check; Lemma 0 gives (H1).

Lemma B.5 (Gershgorin's bound). Let H be an interval matrix whose entries contain those of every symmetric matrix S in a set. If, for every i , the lower end of H_{ii} exceeds the sum over $j \neq i$ of the upper bounds of $|H_{ij}|$, then every such S is positive definite.

Proof. Let $S v = \lambda v$ with v real and nonzero (S is symmetric), and let i be an index with $|v_i|$ maximal. Then $(\lambda - S_{ii}) v_i = \sum_{j \neq i} S_{ij} v_j$, so $\lambda \geq S_{ii} - \sum_{j \neq i} |S_{ij}| |v_j| / |v_i| \geq S_{ii} - \sum_{j \neq i} |S_{ij}| > 0$. QED

`certify_rest_wave.lemma_B` applies Lemma B.5 to $H = D A + A^T D$, with the entries of H evaluated from the interval matrix A that encloses $T_B D f(y, K) T_B^{-1}$ over $B \times [K_1, K_2]$ (for (H2)(i)).

Appendix C. The prior-article search

Dates: 2026-09-25 to 2026-09-27. Sources: arXiv (abstract and all-field search; the API refused requests from our machine), zbMATH Open (web and API), PubMed, Crossref, Semantic Scholar (search, and the citing papers of Hastings 1976, Carpenter 1977 and Arioli and Koch 2015), OpenAlex (its free budget was exhausted), a general web search, and the book of abstracts of Dynamics, Topology and Computations 2025. Query families: "Hodgkin-Huxley" with travelling or traveling wave or pulse, propagated or propagating action potential, homoclinic, existence or cable, and with computer-assisted, rigorous numerics, interval arithmetic or validated; and "computer-assisted" with travelling wave, homoclinic, nerve, excitable or conductance-based. Positive controls: the computer-assisted FitzHugh-Nagumo results (Arioli and Koch 2015; Czechowski and Zgliczynski 2016) were found each time.

Result: no proof, with or without a computer, of the existence of the pulse of the unmodified 1952

equations, and no computer-assisted travelling-wave result for Hodgkin-Huxley or any conductance-based model. Every existence proof found (Hastings 1976; Carpenter 1977; Ikeda, Mimura and Tsujikawa 1987 and 1989, from their abstracts) uses artificial small parameters. Not read: Hastings (1976) beyond pp. 229-230, Foote and Chen (1981), Huxley (1959), and the MathSciNet reviews (not reachable); Google Scholar was not reachable. Hastings (1976) beyond pp. 229-230 and Foote and Chen (1981) could not be obtained. These are the limits of the statement on earlier work in Section 1: we found no earlier proof, within them.

References

- Arioli, G., Koch, H. Existence and stability of traveling pulse solutions of the FitzHugh-Nagumo equation. *Nonlinear Anal.* 113 (2015) 51-70. doi:10.1016/j.na.2014.09.023. (Read in the parts cited; not used by the proof.)
- Carpenter, G. A. Nerve impulse equations. In: *Structural Stability, the Theory of Catastrophes, and Applications in the Sciences*, Lecture Notes in Math. 525, Springer, 1976, 58-76. doi:10.1007/BFb0077843. Zbl 0364.92015. (Listed by zbMATH next to the 1977 paper; not read; not used by the proof.)
- Carpenter, G. A. A geometric approach to singular perturbation problems with applications to nerve impulse equations. *J. Differential Equations* 23 (1977) 335-367. doi:10.1016/0022-0396(77)90116-4. (Read in full; not used by the proof.)
- Conley, C. On traveling wave solutions of nonlinear diffusion equations. In: *Dynamical Systems, Theory and Applications* (J. Moser, ed.), Lecture Notes in Physics 38, Springer, 1975, 498-510. doi:10.1007/3-540-07171-7_13. (Credited for the method; not read; not used by the proof.)
- Conley, C. Isolated Invariant Sets and the Morse Index. *CBMS Regional Conference Series in Mathematics* 38, American Mathematical Society, 1978. Zbl 0397.34056. (Credited for the method; not read; not used by the proof.)
- Czechowski, A., Zgliczynski, P. Existence of periodic solutions of the FitzHugh-Nagumo equations for an explicit range of the small parameter. *SIAM J. Appl. Dyn. Syst.* 15 (2016) 1615-1655. doi:10.1137/15M1007707; arXiv:1502.02451. (Found in the search; not read.)
- FLINT team. FLINT: Fast Library for Number Theory, version 3.6.0 (which contains Arb), <https://flintlib.org>; used through python-flint 0.9.0, <https://github.com/flintlib/python-flint>. (The software of the computations.)
- Foote, J. R., Chen, K.-H. Traveling wave properties of the Hodgkin-Huxley equations. *Chinese J. Math.* 9 (1981) 1-23. Zbl 0472.35048. (Could not be obtained; bears only on the paragraph on earlier work.)
- Hastings, S. P. On travelling wave solutions of the Hodgkin-Huxley equations. *Arch. Rational Mech. Anal.* 60 (1976) 229-257. doi:10.1007/BF01789258. (Read: pp. 229-230; the rest could not be obtained; bears only on the paragraph on earlier work.)
- Hodgkin, A. L., Huxley, A. F. A quantitative description of membrane current and its application to conduction and excitation in nerve. *J. Physiol.* 117 (1952) 500-544. (Read: pp. 519-528 and Table 3.)
- Huxley, A. F. Ion movements during nerve activity. *Ann. N.Y. Acad. Sci.* 81 (1959) 221-246. doi:10.1111/j.1749-6632.1959.tb49311.x. (Not read; cited for the slow pulse, which is not claimed.)
- Ikeda, H., Mimura, M., Tsujikawa, T. Slow traveling wave solutions to the Hodgkin-Huxley equations. In: *Recent Topics in Nonlinear PDE III*, Lecture Notes Numer. Appl. Anal. 9 (1987) 1-73; and *Japan J. Appl. Math.* 6 (1989) 1-66, doi:10.1007/BF03167914. (Abstracts

- read; not used by the proof.)
- Johansson, F. Arb: efficient arbitrary-precision midpoint-radius interval arithmetic. *IEEE Trans. Comput.* 66 (2017) 1281-1292. doi:10.1109/TC.2017.2690633. (The ball arithmetic the computations use.)
 - Lohner, R. J. Einschliessung der Loesung gewoehnlicher Anfangs- und Randwertaufgaben und Anwendungen. Dissertation, Universitaet Karlsruhe, 1988. Zbl 0663.65074. (Credited for the method; not read; not used by the proof.)
 - mpmath development team. mpmath: a Python library for arbitrary-precision floating-point arithmetic, version 1.3.0, <https://mpmath.org>. (The interval arithmetic of the independent block check.)
 - Teschl, G. Ordinary Differential Equations and Dynamical Systems. Graduate Studies in Mathematics 140, American Mathematical Society, 2012. (Read in the author's freely available preliminary version, whose page numbers are given: Corollary 2.15, p. 52; the Routh-Hurwitz criterion, p. 72; Theorem 6.1, p. 189; Lemmas 6.3 and 6.5, p. 193; Lemma 6.6, p. 194; Theorems 9.4 and 9.5, p. 259.)
 - Wazewski, T. Sur un principe topologique de l'examen de l'allure asymptotique des integrales des equations differentielles ordinaires. *Ann. Soc. Polon. Math.* 20 (1947) 279-313 (zbMATH gives 1948: Zbl 0032.35001). (Credited for the method; not read; not used by the proof.)
 - Zgliczynski, P. Covering relations, cone conditions and the stable manifold theorem. *J. Differential Equations* 246 (2009) 1774-1819. doi:10.1016/j.jde.2008.12.019. (Credited for the method; not read; not used by the proof.)